

RW4 Full Modular Run Summary (Combined)

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

steps=22

failed=0

modules_with_outputs=22

indexed_sheets=117

RW4 run-all now includes both farm-gate reconstruction sheets and consolidated robustness outputs.

Table: Run_All_Summary

	step	status	output_dir	error
	sheet_FarmGateUA_initial	ok	/Users/getapple/Documents...	
	sheet_FarmGateUA_filled	ok	/Users/getapple/Documents...	
	sheet_ProducerUA	ok	/Users/getapple/Documents...	
	sheet_ConsumerUA	ok	/Users/getapple/Documents...	
	sheet_EU	ok	/Users/getapple/Documents...	
	sheet_ProZorro	ok	/Users/getapple/Documents...	
	sheet_Silpo	ok	/Users/getapple/Documents...	
	sheet_Novus	ok	/Users/getapple/Documents...	
	sheet_CME	ok	/Users/getapple/Documents...	
model_short_chain_regional		ok	/Users/getapple/Documents...	
	model_ardl	ok	/Users/getapple/Documents...	
	model_ecm	ok	/Users/getapple/Documents...	
	model_nardl	ok	/Users/getapple/Documents...	
	model_vecm	ok	/Users/getapple/Documents...	
	model_discounts	ok	/Users/getapple/Documents...	
model_intersection_bidire...		ok	/Users/getapple/Documents...	
	model_forecast_knn	ok	/Users/getapple/Documents...	
model_secondary_synthetic...		ok	/Users/getapple/Documents...	
	graphs_decomposition	ok	/Users/getapple/Documents...	
	graphs_overlay_ln	ok	/Users/getapple/Documents...	
	graphs_correlations_lags	ok	/Users/getapple/Documents...	
	graphs_brand_region	ok	/Users/getapple/Documents...	

Table: Artifacts_By_Module

module	output_dir	xlsx_count	pdf_count	md_count	png_count	xlsx_files	pdf_files	md_files
graphs_brand_region	/Users/getapple/Documents...	1	1	1	3	graphs_brand_region_outpu...	graphs_brand_region_repor...	graphs_brand_region_repor...
graphs_correlations_lags	/Users/getapple/Documents...	1	1	1	2	graphs_correlations_lags_...	graphs_correlations_lags_...	graphs_correlations_lags_...
graphs_decomposition	/Users/getapple/Documents...	1	1	1	24	graphs_decomposition_outp...	graphs_decomposition_repo...	graphs_decomposition_repo...
graphs_overlay_ln	/Users/getapple/Documents...	1	1	1	21	graphs_overlay_ln_output...	graphs_overlay_ln_report.pdf	graphs_overlay_ln_report.md
model_ardl	/Users/getapple/Documents...	1	1	1	1	model_ardl_output.xlsx	model_ardl_report.pdf	model_ardl_report.md
model_discounts	/Users/getapple/Documents...	1	1	1	3	model_discounts_output.xlsx	model_discounts_report.pdf	model_discounts_report.md
model_ecm	/Users/getapple/Documents...	1	1	1	1	model_ecm_output.xlsx	model_ecm_report.pdf	model_ecm_report.md
model_forecast_knn	/Users/getapple/Documents...	1	1	1	3	model_forecast_knn_output...	model_forecast_knn_report...	model_forecast_knn_report.md
model_intersection_bidire...	/Users/getapple/Documents...	1	1	1	2	model_intersection_bidire...	model_intersection_bidire...	model_intersection_bidire...
model_nardl	/Users/getapple/Documents...	1	1	1	2	model_nardl_output.xlsx	model_nardl_report.pdf	model_nardl_report.md
model_secondary_synthetic...	/Users/getapple/Documents...	1	1	1	0	secondary_synthetic_cons...	secondary_synthetic_cons...	secondary_synthetic_cons...
model_short_chain_regional	/Users/getapple/Documents...	1	1	1	4	primary_chain Consolidate...	primary_chain Consolidate...	primary_chain Consolidate...
model_vecm	/Users/getapple/Documents...	1	1	1	1	model_vecm_output.xlsx	model_vecm_report.pdf	model_vecm_report.md
sheet_CME	/Users/getapple/Documents...	1	1	1	3	sheet_cme_output.xlsx	sheet_cme_report.pdf	sheet_cme_report.md
sheet_ConsumerUA	/Users/getapple/Documents...	1	1	1	3	sheet_consumerua_output.xlsx	sheet_consumerua_report.pdf	sheet_consumerua_report.md
sheet_EU	/Users/getapple/Documents...	1	1	1	3	sheet_eu_output.xlsx	sheet_eu_report.pdf	sheet_eu_report.md
sheet_FarmGateUA_filled	/Users/getapple/Documents...	1	1	1	4	sheet_farmgateua_filled o...	sheet_farmgateua_filled r...	sheet_farmgateua_filled r...
sheet_FarmGateUA_initial	/Users/getapple/Documents...	1	1	1	4	sheet_farmgateua_initial_...	sheet_farmgateua_initial_...	sheet_farmgateua_initial_...
sheet_Novus	/Users/getapple/Documents...	1	1	1	4	sheet_novus_output.xlsx	sheet_novus_report.pdf	sheet_novus_report.md
sheet_ProZorro	/Users/getapple/Documents...	1	1	1	4	sheet_prozorro_output.xlsx	sheet_prozorro_report.pdf	sheet_prozorro_report.md
sheet_ProducerUA	/Users/getapple/Documents...	1	1	1	3	sheet_producerua_output.xlsx	sheet_producerua_report.pdf	sheet_producerua_report.md
sheet_Silpo	/Users/getapple/Documents...	1	1	1	4	sheet_silpo_output.xlsx	sheet_silpo_report.pdf	sheet_silpo_report.md

Table: Sheets_Index

module	xlsx_file	sheet	rows	cols	has_tests_pattern	has_model_results_pattern
graphs_brand_region	graphs_brand_region_outpu...	Brand_Economic_Metrics	1808	10	0	1
graphs_brand_region	graphs_brand_region_outpu...	Brand_IO_Metrics	43	8	0	0
graphs_brand_region	graphs_brand_region_outpu...	Prozorro_ByRegion	147	20	0	0
graphs_correlations_lags	graphs_correlations_lags...	Corr_Matrix	9	10	0	0
graphs_correlations_lags	graphs_correlations_lags...	Correlations	296	10	0	0
graphs_correlations_lags	graphs_correlations_lags...	Lag_Best	31	7	0	0
graphs_correlations_lags	graphs_correlations_lags...	Lag_Profiles	872	7	0	0
graphs_decomposition	graphs_decomposition_outp...	Decomposition_All	25346	9	0	0
graphs_decomposition	graphs_decomposition_outp...	Decomposition_Index	31	4	0	0
graphs_decomposition	graphs_decomposition_outp...	Decomposition_Summary	31	8	0	0
graphs_overlay_ln	graphs_overlay_ln_output....	BeforeAfterLN_All	25452	8	0	0
graphs_overlay_ln	graphs_overlay_ln_output....	BeforeAfterLN_Index	36	4	0	0
graphs_overlay_ln	graphs_overlay_ln_output....	Overlay_All	11867	12	0	0
graphs_overlay_ln	graphs_overlay_ln_output....	Overlay_Index	9	4	0	0
model_ardl	model_ardl_output.xlsx	ARDL_Summary	1383	41	1	1
model_discounts	model_discounts_output.xlsx	Asymmetry_Observed_vs_Bas...	1322	20	0	1
model_discounts	model_discounts_output.xlsx	Discount_Strategy_Synthesis	8	3	0	0
model_discounts	model_discounts_output.xlsx	Promo_State_Depth	1	5	0	0
model_discounts	model_discounts_output.xlsx	Promo_State_Incidence	1	5	0	0
model_discounts	model_discounts_output.xlsx	Promo_State_Type	1	5	0	0
model_discounts	model_discounts_output.xlsx	Silpo_Discounts_Depth	1	5	0	0
model_discounts	model_discounts_output.xlsx	Silpo_Discounts_Occurrence	1	5	0	0
model_discounts	model_discounts_output.xlsx	Silpo_Transmission_PromoCtrl	1322	20	0	1
model_ecm	model_ecm_output.xlsx	ECM_Summary	286	41	1	1
model_forecast_knn	model_forecast_knn_output...	Forecast_Predictions	480	6	0	0
model_forecast_knn	model_forecast_knn_output...	Forecast_Summary	8	7	0	0
model_forecast_knn	model_forecast_knn_output...	Synthetic_Influence_Coeff...	31	7	0	1
model_forecast_knn	model_forecast_knn_output...	Synthetic_Retail_Series	91711	8	0	0
model_forecast_knn	model_forecast_knn_output...	Synthetic_to_Consumer_Link	3	7	0	1
model_forecast_knn	model_forecast_knn_output...	Ultimate_Consumer_Price	143	5	0	0
model_intersection_bidire...	model_intersection_bidire...	Bidirectional_Granger	1	1	0	0
model_intersection_bidire...	model_intersection_bidire...	Bidirectional_Results	1	1	0	0
model_intersection_bidire...	model_intersection_bidire...	CrossTable_Correlations	68	10	0	0
model_intersection_bidire...	model_intersection_bidire...	Intersection_Combination_...	0	0	0	0
model_intersection_bidire...	model_intersection_bidire...	Intersection_Combination_...	1	1	0	0
model_nardl	model_nardl_output.xlsx	NARDL_Summary	3028	41	1	1
model_secondary_synthetic...	secondary_synthetic_cons...	Synthetic_Consumer_Link	3	8	0	1
model_secondary_synthetic...	secondary_synthetic_cons...	Synthetic_Consumer_Predic...	166	9	0	0
model_short_chain_regional	primary_chain_consolidate...	AveragePrice_Chain_Transm...	749	41	1	1
model_short_chain_regional	primary_chain_consolidate...	Benchmark_Comparison	1992	18	0	0

Table: Tests Interpretation

module	xlsx_file	sheet	n_rows	il_like_share	cointegration_support_share	interpretation
model_ardl	model_ardl_output.xlsx	ARDL_Summary	1383	0.0	0.0	I(1)-like share=0.00; coi...
model_ecm	model_ecm_output.xlsx	ECM_Summary	286	0.0	0.0	I(1)-like share=0.00; coi...
model_nardl	model_nardl_output.xlsx	NARDL_Summary	3028	0.0	0.0	I(1)-like share=0.00; coi...
model_short_chain_regional	primary_chain_consolidate...	AveragePrice_Chain Transm...	749	0.0	0.0	I(1)-like share=0.00; coi...
model_short_chain_regional	primary_chain_consolidate...	Consolidated_ModelCoeffic...	10141	0.0	0.0	I(1)-like share=0.00; coi...
model_short_chain_regional	primary_chain_consolidate...	Consolidated PreTests	4312	0.0	0.5051020408163265	I(1)-like share=0.00; coi...
model_short_chain_regional	primary_chain_consolidate...	RawMilk_To_Product_Transm...	3465	0.0	0.0	I(1)-like share=0.00; coi...
model_short_chain_regional	primary_chain_consolidate...	Retailer_Brand_Transmission	962	0.0	0.0	I(1)-like share=0.00; coi...
model_short_chain_regional	primary_chain_consolidate...	ReverseFlow_ModelCoeffici...	4548	0.0	0.0	I(1)-like share=0.00; coi...
model_vecm	model_vecm_output.xlsx	VECM_Summary	288	0.0	0.0	I(1)-like share=0.00; coi...
sheet_CME	sheet_cme_output.xlsx	tests	3	0.3333333333333333	0.0	I(1)-like share=0.33; coi...
sheet_ConsumerUA	sheet_consumerua_output.xlsx	tests	9	1.0	0.0	I(1)-like share=1.00; coi...
sheet_EU	sheet_eu_output.xlsx	tests	15	0.3333333333333333	0.0	I(1)-like share=0.33; coi...
sheet_FarmGateUA_filled	sheet_farmgateua_filled o...	tests	3	1.0	0.0	I(1)-like share=1.00; coi...
sheet_FarmGateUA_initial	sheet_farmgateua_initial_...	tests	3	1.0	0.0	I(1)-like share=1.00; coi...
sheet_Novus	sheet_novus_output.xlsx	tests	27	0.0	0.0	I(1)-like share=0.00; coi...
sheet_ProZorro	sheet_prozorro_output.xlsx	tests	18	0.2222222222222222	0.0	I(1)-like share=0.22; coi...
sheet_ProducerUA	sheet_producerua_output.xlsx	tests	15	1.0	0.0	I(1)-like share=1.00; coi...
sheet_Silpo	sheet_silpo_output.xlsx	tests	24	0.0	0.0	I(1)-like share=0.00; coi...

Table: Results_Interpretation

module	xlsx_file	sheet	n_rows	robust_across_reconstruction_share	core_finding_share	interpretation
graphs_brand_region	graphs_brand_region_outpu...	Brand_Economic_Metrics	1808	nan	nan	mean coef =27.5983 sig...
model_ardl	model_ardl_output.xlsx	ARDL_Summary	1383	0.0	0.0	mean coef =2.6492 sign...
model_discounts	model_discounts_output.xlsx	Asymmetry_Observed_vs_Bas...	1322	nan	nan	mean coef =3.4146
model_discounts	model_discounts_output.xlsx	Silpo_Transmission_PromoCtrl	1322	nan	nan	mean coef =3.4146
model_ecm	model_ecm_output.xlsx	ECM_Summary	286	0.8951048951048951	0.10139860139860139	mean coef =0.7494 sign...
model_forecast_knn	model_forecast_knn_output...	Synthetic_Influence_Coeff...	31	nan	nan	mean coef =0.5000
model_forecast_knn	model_forecast_knn_output...	Synthetic_to_Consumer_Link	3	nan	nan	mean coef =0.8160 sign...
model_nardl	model_nardl_output.xlsx	NARDL_Summary	3028	0.8127476882430648	0.2807133421400264	mean coef =4.1309 sign...
model_secondary_synthetic...	secondary_synthetic_cons...	Synthetic_Consumer_Link	3	nan	nan	mean coef =0.4950 sign...
model_short_chain_regional	primary_chain Consolidate...	AveragePrice_Chain Transm...	749	0.2590120160213618	0.11214953271028037	mean coef =2.9094 sign...
model_short_chain_regional	primary_chain Consolidate...	Consolidated_ModelCoeffici...	10141	0.26792229563159453	0.08667784242185189	mean coef =3.5293 sign...
model_short_chain_regional	primary_chain Consolidate...	Consolidated_PreTests	4312	nan	nan	significance share (p<0.0...
model_short_chain_regional	primary_chain Consolidate...	FarmGate_Direct_Summary	163	nan	nan	mean coef =8.5959
model_short_chain_regional	primary_chain Consolidate...	FarmGate_Reverse_Summary	146	nan	nan	mean coef =0.1189
model_short_chain_regional	primary_chain Consolidate...	FarmGate_Source_Comparison	2379	0.48886086591004624	nan	robust-across-reconstruct...
model_short_chain_regional	primary_chain Consolidate...	RawMilk_To_Product Transm...	3465	0.23694083694083695	0.02655122655122655	mean coef =8.1330 sign...
model_short_chain_regional	primary_chain Consolidate...	Retailer_Brand_Transmission	962	0.12681912681912683	0.057172557172557176	mean coef =8.4973 sign...
model_short_chain_regional	primary_chain Consolidate...	ReverseFlow_ModelCoeffici...	4548	0.28979771328056286	0.10180299032541777	mean coef =0.3376 sign...
model_vecm	model_vecm_output.xlsx	VECM_Summary	288	0.0	0.0	mean coef =11.6733 sig...
sheet_CME	sheet_cme_output.xlsx	tests	3	nan	nan	significance share (p<0.0...
sheet_ConsumerUA	sheet_consumerua_output.xlsx	tests	9	nan	nan	significance share (p<0.0...
sheet_EU	sheet_eu_output.xlsx	tests	15	nan	nan	significance share (p<0.0...
sheet_FarmGateUA_filled	sheet_farmgateua_filled o...	tests	3	nan	nan	significance share (p<0.0...
sheet_FarmGateUA_initial	sheet_farmgateua_initial...	tests	3	nan	nan	significance share (p<0.0...
sheet_Novus	sheet_novus_output.xlsx	tests	27	nan	nan	Result table has no stand...
sheet_ProZorro	sheet_prozorro_output.xlsx	tests	18	nan	nan	significance share (p<0.0...
sheet_ProducerUA	sheet_producerua_output.xlsx	tests	15	nan	nan	significance share (p<0.0...
sheet_Silpo	sheet_silpo_output.xlsx	tests	24	nan	nan	Result table has no stand...

Table: Module_Block_Interpretation

module	output_dir	xlsx_files	pdf_files	md_files	png_count	interpretation
graphs_brand_region	/Users/getapple/Documents...	graphs_brand_region_outpu...	graphs_brand_region_repor...	graphs_brand_region_repor...	3	mean coef =27.5983 sig...
graphs_correlations_lags	/Users/getapple/Documents...	graphs_correlations_lags_...	graphs_correlations_lags_...	graphs_correlations_lags_...	2	No explicit test/model ta...
graphs_decomposition	/Users/getapple/Documents...	graphs_decomposition_outp...	graphs_decomposition_repo...	graphs_decomposition_repo...	24	No explicit test/model ta...
graphs_overlay_ln	/Users/getapple/Documents...	graphs_overlay_ln_output...	graphs_overlay_ln_report.pdf	graphs_overlay_ln_report.md	21	No explicit test/model ta...
model_ardl	/Users/getapple/Documents...	model_ardl_output.xlsx	model_ardl_report.pdf	model_ardl_report.md	1	I(1)-like share=0.00; coi...
model_discounts	/Users/getapple/Documents...	model_discounts_output.xlsx	model_discounts_report.pdf	model_discounts_report.md	3	mean coef =3.4146
model_ecm	/Users/getapple/Documents...	model_ecm_output.xlsx	model_ecm_report.pdf	model_ecm_report.md	1	I(1)-like share=0.00; coi...
model_forecast_knn	/Users/getapple/Documents...	model_forecast_knn_output...	model_forecast_knn_report...	model_forecast_knn_report.md	3	mean coef =0.5000
model_intersection_bidire...	/Users/getapple/Documents...	model_intersection_bidire...	model_intersection_bidire...	model_intersection_bidire...	2	No explicit test/model ta...
model_nardl	/Users/getapple/Documents...	model_nardl_output.xlsx	model_nardl_report.pdf	model_nardl_report.md	2	I(1)-like share=0.00; coi...
model_secondary_synthetic...	/Users/getapple/Documents...	secondary_synthetic_cons...	secondary_synthetic_cons...	secondary_synthetic_cons...	0	mean coef =0.4950 sign...
model_short_chain_regional	/Users/getapple/Documents...	primary_chain_consolidate...	primary_chain_consolidate...	primary_chain_consolidate...	4	I(1)-like share=0.00; coi...
model_vecm	/Users/getapple/Documents...	model_vecm_output.xlsx	model_vecm_report.pdf	model_vecm_report.md	1	I(1)-like share=0.00; coi...
sheet_CME	/Users/getapple/Documents...	sheet_cme_output.xlsx	sheet_cme_report.pdf	sheet_cme_report.md	3	I(1)-like share=0.33; coi...
sheet_ConsumerUA	/Users/getapple/Documents...	sheet_consumerua_output.xlsx	sheet_consumerua_report.pdf	sheet_consumerua_report.md	3	I(1)-like share=1.00; coi...
sheet_EU	/Users/getapple/Documents...	sheet_eu_output.xlsx	sheet_eu_report.pdf	sheet_eu_report.md	3	I(1)-like share=0.33; coi...
sheet_FarmGateUA_filled	/Users/getapple/Documents...	sheet_farmgateua_filled o...	sheet_farmgateua_filled r...	sheet_farmgateua_filled r...	4	I(1)-like share=1.00; coi...
sheet_FarmGateUA_initial	/Users/getapple/Documents...	sheet_farmgateua_initial_...	sheet_farmgateua_initial_...	sheet_farmgateua_initial_...	4	I(1)-like share=1.00; coi...
sheet_Novus	/Users/getapple/Documents...	sheet_novus_output.xlsx	sheet_novus_report.pdf	sheet_novus_report.md	4	I(1)-like share=0.00; coi...
sheet_ProZorro	/Users/getapple/Documents...	sheet_prozorro_output.xlsx	sheet_prozorro_report.pdf	sheet_prozorro_report.md	4	I(1)-like share=0.22; coi...
sheet_ProducerUA	/Users/getapple/Documents...	sheet_producerua_output.xlsx	sheet_producerua_report.pdf	sheet_producerua_report.md	3	I(1)-like share=1.00; coi...
sheet_Silpo	/Users/getapple/Documents...	sheet_silpo_output.xlsx	sheet_silpo_report.pdf	sheet_silpo_report.md	4	I(1)-like share=0.00; coi...